

MOHITH D K

Bangalore, Karnataka | P: 9019325337 | mohithog7@gmail.com | [linkedin](#) | [Github](#)

EDUCATION

PES UNIVERSITY

B.tech in Computer science engineering(AI/ML)

Bangalore, Karnataka

Aug 2022 - May 2026

- Relevant Coursework: Advanced Data Analytics, Deep Learning, LLM, NLP, Machine Learning, Computer Networks, Operating Systems, Cloud Computing .

EXPERIENCE

A P MOLLER MAERSK

Jun 2025 - Jul 2025

AI Intern

Remote

- Developed Conversational AI and sentiment-driven classification modules by fine-tuning BERT for customer feedback analysis, enabling intelligent ticket handling and smoother customer-support workflows.
- Fine-tuned a BERT-based NLP pipeline (F1 : 0.91) to automate sentiment and priority detection in customer tickets, streamlining support operations and enhancing scalability for high-volume service environments.

SpectoV

Jun 2025 - Jul 2025

AI Intern

Remote

- Developed Conversational AI and sentiment-driven classification modules by fine-tuning BERT for customer feedback analysis, enabling intelligent ticket handling and smoother customer-support workflows.
- Fine-tuned a BERT-based NLP pipeline (F1 : 0.91) to automate sentiment and priority detection in customer tickets, streamlining support operations and enhancing scalability for high-volume service environments.

RAPID Research Lab

May 2025 - Jun 2025

Research Intern

on-site

- Conducted cutting-edge research on "Temporal Belief Dynamics in BERT", employing a graph-theoretic approach to analyze interpretable representation drift for a publishable AI/ML project.
- Developed and refined novel machine learning models and methodologies within a research-driven environment, contributing to trustworthy AI techniques that strengthen model transparency and performance monitoring.

PROJECTS

Temporal Belief Dynamics in BERT – Research Project

Jun 2025 - Jul 2025

- Mapped the evolution of BERT's internal "beliefs" across 10 fine-tuning epochs, using graph-theoretic networks and 2D embeddings to uncover and visualize hidden representation drift with research-grade clarity.
- Shared these findings in a research paper that was accepted for publication at the 7th International Conference on Communication and Intelligent Systems 2025.

ArsenicCure - AI-Powered Arsenic Skin Lesion Detection and Severity Grading

Sep 2024 - present

- Developed a hybrid deep learning model integrating ResNet-18, MobileNet-v2, and ViT architectures using PyTorch, achieving 88.33% test accuracy and an F1-score of 0.88 on a balanced dataset of 3,462 skin lesion images classified into mild, moderate, normal, and severe categories.
- Currently deploying a full-stack application for real-time severity grading and clinician validation, building upon previously implemented explainable AI features (Grad-CAM) and a tech stack including FastAPI, Docker, and MongoDB.

Advanced Multimodal Deep Learning Framework for Proactive Cyberbullying Identification

Oct 2024

- Collaborated on the development of a multimodal deep learning model integrating LSTM for text and ResNet50 for image features to predict cyberbullying behaviors, achieving 85% detection accuracy.
- Actively participated in data preprocessing (Multi-Modal Hate Speech Dataset) and model optimization (dropout layers) leveraging Python with Keras/TensorFlow.

PUBLICATIONS

M.D.K. et al, "*Temporal Belief Dynamics in BERT: A Graph-Theoretic Approach to Interpretable Representation Drift*," published at 7th International Conference on Communication and Intelligent Systems (ICCIS 2025), BITS Pilani, Goa, *Springer LNCS*

TECHNICAL SKILLS

Languages & Core Development: Python, SQL, HTML/CSS, Reactjs

AI/ML & Data: PyTorch, Keras, TensorFlow, Hugging Face, Pandas, LLM, NLP, RAG, LangChain

Backend, Databases & Streaming: FastAPI, MongoDB, Nodejs, MySQL, Apache Kafka

DevOps & Tools: Docker, Kubernetes, Git, Jupyter Notebooks, VS Code, Wireshark

ACHIEVEMENTS

Winners: HACK-AI 24-hour Data Analytics Hackathon - PES University in association with Vision Karnataka Foundation.